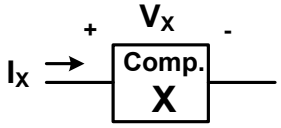
ECE245 Helpful Equations Sheet 1 (Version Fall 2020)

$i = \frac{dq}{dt}$ (amp = coulombs/sec) $v = \frac{dW}{dq} = iR$ (volt = joules/coulomb)

$R = \frac{\rho L}{A} = \frac{v}{i}$ (ohm = volt/amp) $p = \frac{dW}{dt} = vi = \frac{v^2}{R} = i^2 R$ (watt = joules/sec)

$E = \int p dt$ (Energy in joules or kWh)



Passive Sign Convention $P_X = V_X I_X$ absorbed by "Comp. X"

Kirchhoff's Laws: KVL: $\sum_{Loop} V = 0$ KCL: $\sum_{node} I = 0$

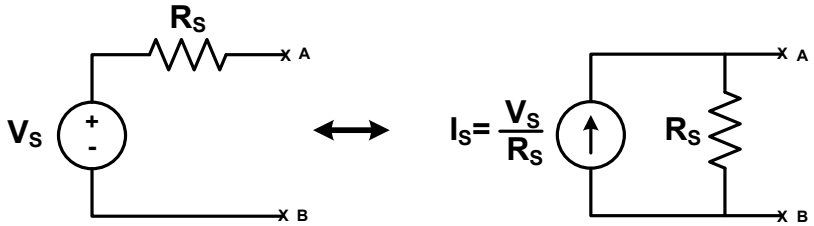
Resistor Comb.: **SERIES** **PARALLEL**

$$R_{eq} = R_1 + R_2 + R_3 + \ldots \quad R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots} \qquad R_{2inParallel} = \frac{R_1 R_2}{R_1 + R_2}$$

Voltage Divider: $V_X = \frac{R_X}{\sum_{series} R} \times V_{series}$ **Current Divider**: $I_X = \frac{R_{eq,parallel}}{R_X} \times I_{TOT,par}$

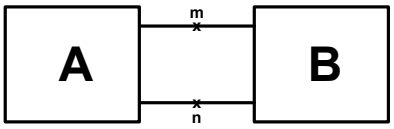
Nodal Analysis: (KCL) $I_{out,X} = \frac{V_X - V_Y}{R_{XY}}$ **Mesh Analysis**: (KVL) $V_{12} = (I_1 - I_2) R_{12}$

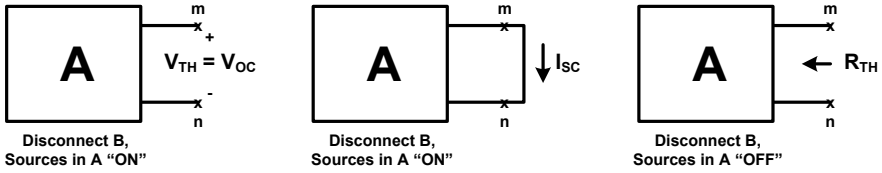
Source Transformation:

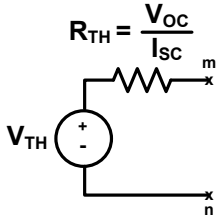


Superposition: $V_X = V_{X1} + V_{X2} + V_{X3} + \ldots = V_X\big|_{\substack{S1,ON \\ S2,S3\ldots OFF}} + V_X\big|_{\substack{S2,ON \\ S1,S3\ldots OFF}} + V_X\big|_{\substack{S3,ON \\ S1,S2\ldots OFF}} + \ldots$

Thevenin Equivalent: Given Networks A & B, the Thevenin of Network A is found from







Maximum Power Transfer: Set $R_{Load} = R_{out} = R_{TH}$ $\quad P_{Load} = I_{Load}^2 R_{Load} = \frac{V_{Load}^2}{R_{Load}}$

Metric Prefixes

Giga	G	10^9
Mega	M	10^6
Kilo	k	10^3
Milli	m	10^{-3}
Micro	μ	10^{-6}
Nano	n	10^{-9}
Pico	p	10^{-12}

Resistor Color Code **4th Band**: Gold = 5%; Silver = 10%; None = 20%

Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Gray	White
0	1	2	3	4	5	6	7	8	9

Resistor Wattage/Size

1/8W	1/4W	1/2W	1W	2W
3.3mm	6.0mm	8.5mm	11mm	15mm